

CCTV Surveillance System Assessment Report

Existing System Assessment and Upgrade Recommendations

Executive Summary

A comprehensive technical assessment of the existing CCTV surveillance system was conducted following an on-site inspection. The evaluation focused on system architecture, camera coverage, equipment condition, recording capability, and operational performance. The purpose of this assessment is to identify current system limitations, evaluate operational reliability, and recommend engineering improvements that will enhance security, surveillance effectiveness, and future system scalability.

1. Assessment Scope

- Evaluation of installed surveillance cameras.
- Verification of camera coverage and monitoring effectiveness.
- Inspection of recording equipment and storage capacity.
- Assessment of system health and operational condition.
- Identification of maintenance requirements.
- Development of recommendations for future expansion and system optimization.

2. Existing CCTV Infrastructure

The current surveillance system consists of three (3) Hikvision outdoor Bullet cameras equipped with infrared day/night functionality to provide continuous monitoring under varying lighting conditions.

- Camera 01 – Monitoring the internal courtyard from the eastern side.
- Camera 02 – Covering the eastern external façade of the building.
- Camera 03 – Covering the western external façade of the building.
- Continuous 24/7 surveillance coverage.

3. Technical Findings

Camera Condition

Preventive maintenance is recommended, including lens cleaning to improve image clarity and reduce visual degradation.

Signal Transmission

A damaged BNC connector was identified and should be replaced to restore signal integrity and eliminate the risk of intermittent video loss.

Recording Equipment

The system utilizes a Hikvision 4-channel DVR. Three channels are currently in use, while one channel remains available for future expansion. Recording retention can be extended through storage upgrades and recording parameter optimization.

4. System Assessment

The existing surveillance system is operational and capable of supporting the site's current security requirements. However, several engineering improvements are recommended to enhance system performance, reliability, and future scalability.

5. Engineering Recommendations

Install an Additional Entrance Camera

Install an additional surveillance camera at the main internal entrance to improve access monitoring and strengthen security.

Enable Audio Recording

Enable audio recording where operationally required and legally permitted, in accordance with applicable regulations and organizational policies.

Increase Recording Retention

Extend recording duration by upgrading HDD storage capacity and optimizing recording bitrate and resolution.

6. Client Information Required

Before preparing the final implementation plan, the following information is required:

- Purpose of audio recording (Security, Operational, Documentation, Training).
- Required recording schedule (24/7 or operational hours only).

Conclusion

The proposed recommendations will improve surveillance coverage, operational reliability, recording capability, and future system scalability. A detailed implementation plan will be prepared after client approval and confirmation of project requirements.

Prepared By

Afnan Saleh Alroaished

Project Manager